

第三章：分离变量法和特殊函数

§3.1 齐次方程、齐次边值条件的定问题

齐次边值的有界弦自由振动：

$$\begin{cases} \text{初值条件} & \begin{cases} \partial_t^2 u = a^2 \partial_x^2 u & , (x, t) \in [0, l] \times [0, \infty) \\ u(x, 0) = \varphi & , \partial_t u(x, 0) = \psi & , x \in [0, l] \end{cases} \\ \text{边值条件} & \begin{cases} u(0, t) = 0, & u(l, t) = 0 & , t \in [0, \infty) \end{cases} \end{cases}$$

Idea

Find solution in the form

$$u(x, t) = X(x) T(t) \quad \leftarrow u = \sum_n X_n(x) T_n(t) \quad \text{fixed } n$$

$$X_n T_n'' = a^2 X_n'' T_n \quad \mapsto \quad \frac{X_n''}{X_n} = \frac{T_n''}{a^2 T_n} \quad \text{当且仅当} = -\lambda_n \text{ (cons.)}$$

$$\begin{cases} X_n'' + \lambda_n X_n = 0 & T_n'' + a^2 \lambda_n T_n = 0 \end{cases}$$

$$\begin{cases} X_n(0) = X_n(l) = 0 & \mapsto \sum_n X_n(x) T_n(0) = \varphi(x), \sum_n X_n(x) T_n'(0) = \psi(x) \end{cases}$$

需满足某种线性性 (Fourier 级数) \$T_n(0)\$ \$T_n'(0)\$

$$\begin{cases} X'' + \lambda X = 0 & , 0 \leq x \leq l \\ X(0) = X(l) = 0 \end{cases}$$

Find non-trivial solution \$X(x) \neq 0\$

$$\begin{aligned} \boxed{\lambda > 0}, & X = C_1 \sin \sqrt{\lambda} x + C_2 \cos \sqrt{\lambda} x \Rightarrow C_2 = 0, \sin \sqrt{\lambda} l = 0, \sqrt{\lambda} l = n\pi, \lambda = \frac{n^2 \pi^2}{l^2} \quad (n=1, 2, 3, \dots) \\ \lambda = 0, & X = C_1 x + C_2 \rightarrow X = 0; \\ \lambda < 0, & X = C_1 e^{\sqrt{\lambda} x} + C_2 e^{-\sqrt{\lambda} x} \rightarrow X = 0. \end{aligned}$$

特征值: $\lambda_n = \frac{n^2 \pi^2}{l^2}$, 特征函数: $X_n(x) = \sin \frac{n\pi}{l} x, n=1, 2, 3, \dots$

正交性: $\int_0^l X_n X_m = 0$ if $n \neq m$

证明: Let X_1, X_2 be eigenfunction of λ_1, λ_2 ($\lambda_1 \neq \lambda_2$).

$$\begin{aligned} \lambda_1 \int_0^l X_1 X_2 &= - \int_0^l X_1'' X_2 = - \int_0^l X_2 dX_1' = -X_2 X_1' \Big|_0^l + \int_0^l X_2' dX_1 \\ &= (-X_2 X_1' + X_2' X_1) \Big|_0^l - \int_0^l X_1 X_2'' = \lambda_2 \int_0^l X_1 X_2 \end{aligned}$$

$$\sum_{m=1}^{\infty} \int_0^L X_n X_m T_m(0) dx = \int_0^L \psi(x) \cdot X_n(x) dx \quad \leftarrow \boxed{T_n(0) = \frac{2}{L} \int_0^L \psi(x) \sin \frac{n\pi}{L} x dx}$$

$$= \sum_{m=1}^{\infty} T_m(0) \int_0^L X_n X_m dx = T_n(0) \int_0^L X_n^2 dx = \frac{L}{2} \cdot T_n(0).$$

$$\int_0^L X_n \cdot \psi(x) dx = \sum_{m=1}^{\infty} \int_0^L X_n X_m T_m'(0) dx = T_n'(0) \cdot \int_0^L X_n^2 dx$$

$$\Rightarrow \boxed{T_n'(0) = \frac{2}{L} \int_0^L \psi(x) \cdot \sin \frac{n\pi}{L} x dx}$$

ODE: $T_n = A_n \cos \frac{an\pi}{L} t + B_n \sin \frac{an\pi}{L} t$

$$A_n = T_n(0) = \frac{2}{L} \int_0^L \psi(x) \sin \frac{n\pi}{L} x dx$$

$$B_n \cdot \frac{an\pi}{L} = T_n'(0) = \frac{2}{L} \int_0^L \psi(x) \sin \frac{n\pi}{L} x dx.$$

eg1. $\begin{cases} \partial_t^2 u = 4x^2 u, & (x, t) \in (0, 2\pi) \times (0, \infty) \\ u(x, 0) = -10 \cos \frac{x}{4}, & \partial_t u(x, 0) = 0, \quad x \in (0, 2\pi) \\ u_x(0, t) = u_x(2\pi, t) = 0, & t \in (0, \infty) \end{cases}$

零边值
永远是正的

step 1: 特征值问题

$$\begin{cases} X''(x) + \lambda X(x) = 0 & (0 < x < 2\pi) \\ X'(0) = X'(2\pi) = 0 \end{cases}$$

$$\lambda_n = \left(\frac{2n-1}{4}\right)^2, \quad X_n = \cos \frac{2n-1}{4} x \quad n=1, 2, 3, \dots$$

step 2: 设 $u(x, t) = \sum_n X_n(x) T_n(t)$

$$\text{则 } T_n''(t) + 4\lambda_n T_n(t) = 0.$$

$$T_n(t) = A_n \cos \frac{2n-1}{2} t + B_n \sin \frac{2n-1}{2} t, \quad n=1, 2, 3, \dots$$

$$u(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{2n-1}{2} t + B_n \sin \frac{2n-1}{2} t \right) \cos \frac{2n-1}{4} x$$

Step 3: $-10 \cos \frac{x}{4} = \sum_{n=1}^{\infty} A_n \cos \frac{2n-1}{4} x, \quad 0 < x < 2\pi$

不需要算积分.

$$0 = \sum_{n=1}^{\infty} \frac{2n-1}{2} B_n \cos \frac{2n-1}{4} x, \quad 0 < x < 2\pi$$

由正交性, 比较系数, $A_1 = -10, A_n = 0 \ (n \geq 2)$
 $B_n = 0$

$$\therefore u(x,t) = -10 \cos \frac{t}{2} \cos \frac{x}{4}$$

齐次边值, 无热源的有界杆热传导

长度 L , 侧面绝热, 一端温度为 0, 另一端绝热的有界杆.

$$\begin{cases} \partial_t u = a^2 \partial_x^2 u & , (x,t) \in (0,L) \times (0,\infty) \\ u(x,0) = \phi(x) & , x \in (0,L) \\ u(0,t) = 0, u_x(L,t) = 0 & , t \in (0,\infty) \end{cases}$$

$$u = \sum X(x) T(t) \quad X T \text{ satisfies } \begin{cases} u_t = a^2 u_{xx} \\ u(0,t) = u_x(L,t) = 0 \\ \text{"} \quad \text{"} \\ X(0) T(t) \quad X'(L) T(t) \end{cases}$$

$$X T' = a^2 X'' T$$

$$\hookrightarrow \frac{X''}{X} = \frac{T'}{a^2 T} = -\lambda$$

Step 1 $X'' + \lambda X = 0$

Find non-trivial solution.

$$X(0) = X'(L) = 0$$

$$\lambda > 0 \quad X = C_1 \cos \sqrt{\lambda} x + C_2 \sin \sqrt{\lambda} x \hookrightarrow \cos \sqrt{\lambda} L = 0$$

step 2

$$\sqrt{\lambda} L = \frac{\pi}{2} + (n-1)\pi, \quad \boxed{\lambda_n = \frac{(n-\frac{1}{2})^2 \pi^2}{L^2}, \quad X_n = \sin \frac{(n-\frac{1}{2})\pi}{L} x, \quad n=1, 2, 3, \dots}$$

$$\sum_{m=1}^{\infty} X_m T_m' = a^2 \sum_{m=1}^{\infty} X_m'' T_m$$

正交: $\int_0^L X_m X_n = 0$ if $m \neq n$.

$$\sum_{m=1}^{\infty} \int_0^L X_n X_m T_m' = -\lambda_n a^2 \sum_{m=1}^{\infty} \int_0^L X_n X_m'' T_m$$

"

$$\cancel{\int X_n^2 T_n'} = -\lambda_n a^2 (\cancel{\int X_n^2}) T_n$$

step 3

$$\Rightarrow \boxed{T_n' + a^2 \lambda_n T_n = 0}$$

$$T_n(t) = A_n \cdot e^{-a^2 \lambda_n t} = A_n \cdot e^{-\frac{(n-\frac{1}{2})^2 \pi^2 a^2}{L^2} t}, \quad n=1, 2, 3, \dots$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} A_n e^{-\frac{(n-\frac{1}{2})^2 \pi^2 a^2}{L^2} t} \sin \frac{(n-\frac{1}{2})\pi}{L} x$$

$$\text{At } t=0, \quad \sum_{n=1}^{\infty} A_n \sin \frac{(n-\frac{1}{2})\pi}{L} x = \phi(x)$$

Fix n

$$\int_0^L \phi X_n = \sum_{m=1}^{\infty} A_m \int_0^L X_m X_n = A_n \int_0^L X_n^2$$

$\xrightarrow{\text{orthogonality}} \quad \xrightarrow{m=n}$

$$\Rightarrow A_n = \frac{2}{L} \int_0^L \phi \cdot \sin \frac{(n-\frac{1}{2})\pi x}{L} dx \quad \leftarrow \frac{L}{2}$$

eg. 求解 Laplace 方程: 分离变量法

$$\Delta u = 0 \quad \begin{cases} u_{xx} + u_{yy} = 0 & (x, y) \in (0, \pi)^2 \\ u(0, y) = 0 \\ u(\pi, y) = \cos^2 y \\ u_y(x, 0) = u_y(x, \pi) = 0 \end{cases}$$

四个边值条件

$$X''Y + XY'' = 0 \Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = \lambda \Rightarrow \begin{cases} X'' - \lambda X = 0 \\ Y'' + \lambda Y = 0 \end{cases}$$

$$u_y = 0 \quad X Y'(0) = X Y'(\pi) = 0 \Rightarrow \begin{cases} Y'' + \lambda Y = 0 & 0 \leq y \leq \pi \\ Y'(0) = Y'(\pi) = 0 \end{cases}$$

此时 λ 可以为 0, Y 取非零常数.

$$\Rightarrow \lambda_n = n^2, \quad Y = \cos ny \quad (n=0, 1, 2, \dots)$$

$$\text{由 } X_n''(x) - \lambda_n X_n(x) = 0$$

$$\Rightarrow X_0(x) = A_0 x + B_0; \quad X_n(x) = A_n e^{-nx} + B_n e^{nx}, \quad (n=1, 2, 3, \dots)$$

$$u(x, y) = A_0 x + B_0 + \sum_{n=1}^{\infty} (A_n e^{-nx} + B_n e^{nx}) \cos ny.$$

$$u(0, y) = 0 \Rightarrow B_0 + \sum_{n=1}^{\infty} (A_n + B_n) \cos ny = 0$$

$$u(\pi, y) = 0 \Rightarrow A_0 \pi + B_0 + \sum_{n=1}^{\infty} (A_n e^{-n\pi} + B_n e^{n\pi}) \cos ny = \cos^2 y = \frac{1}{2} + \frac{1}{2} \cos 2y$$

由正交性.

比较系数: $B_0 = 0, \quad A_n + B_n = 0$.

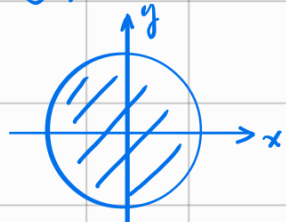
$$A_0 \pi = \frac{1}{2}, \quad A_2 e^{-2\pi} + B_2 e^{2\pi} = \frac{1}{2}$$

$$A_0 = \frac{1}{2\pi}, \quad A_2 = -B_2$$

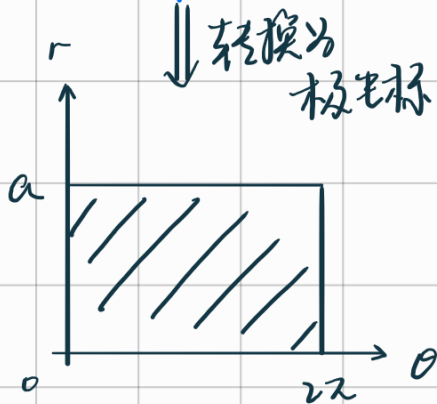
eg. 圆盘上周期性条件的 Laplace 方程:

半径为 a 的薄圆盘: $\bar{Z}_a = \{X = (x, y) \in \mathbb{R}^2: \|X\| \leq a\}$

两面绝热, 边缘温度 $\phi(X)$. 内部无热源.



$$\begin{cases} \Delta u = 0, & X \in \bar{Z}_a \\ u(X) = \phi(X) & X \in S_a := \partial \bar{Z}_a \end{cases}$$



转换为极坐标

$$\bar{u}(r, \theta) = u(r \cos \theta, r \sin \theta)$$

$$\bar{u}_r = u_x \cdot \cos \theta + u_y \sin \theta$$

$$\bar{u}_\theta = u_x (-r \sin \theta) + u_y (r \cos \theta)$$

$$\begin{aligned} \bar{u}_{rr} &= \cos \theta \cdot (u_{xx} \cdot \cos \theta + u_{xy} \sin \theta) \\ &\quad + \sin \theta (u_{xy} \cdot \cos \theta + u_{yy} \sin \theta) \end{aligned}$$

$$= u_{xx} \cdot \cos^2 \theta + u_{xy} \cdot 2 \sin \theta \cos \theta + u_{yy} \cdot \sin^2 \theta$$

$$\begin{aligned} \bar{u}_{\theta\theta} &= (-r \sin \theta) [u_{xx} (-r \sin \theta) + u_{xy} (r \cos \theta)] + \cos \theta \cdot (-r u_x) \\ &\quad + (r \cos \theta) [u_{xy} (-r \sin \theta) + u_{yy} (r \cos \theta)] + (-\sin \theta) \cdot r u_y \\ &= u_{xx} \cdot r^2 \sin^2 \theta + u_{xy} (-2r^2 \sin \theta \cos \theta) + u_{yy} (r^2 \cos^2 \theta) - r \cdot (u_x \cos \theta + u_y \sin \theta) \end{aligned}$$

$$\therefore r^2 \cdot \bar{u}_{rr} + \bar{u}_{\theta\theta} = (u_{xx} + u_{yy}) \cdot r^2 - r \cdot \bar{u}_r = -r \cdot \bar{u}_r$$

$$\therefore r^2 \cdot \bar{u}_{rr} + r \cdot \bar{u}_r + \bar{u}_{\theta\theta} = 0$$

$X = (x, y)$

$$\bar{u}(a, \theta) = \bar{\phi}(\theta) : \phi(a \cos \theta, a \sin \theta), \quad \theta \in [0, 2\pi].$$

加一项: $\bar{u}(r, \theta) = \bar{u}(r, \theta + 2\pi)$ 则是周期的. $(r, \theta) \in [0, a) \times [0, 2\pi)$.

$$\bar{u} = \sum_n R_n(r) H_n(\theta) \quad \leftarrow \boxed{R(r) H(\theta)}$$

$$\frac{r^2 \cdot R'' \cdot H + r \cdot R' \cdot H + R \cdot H''}{R \cdot H} = 0$$

$$\Rightarrow \frac{H''}{H} = -\frac{r^2 R''}{R} - \frac{r R'}{R} = -\lambda$$

step 1 $\begin{cases} H'' + \lambda H = 0 \\ H(\theta) = H(\theta + 2\pi) \end{cases}$

$\lambda < 0$, $H(\theta)$ 为指数函数, 不满足周期性;

$\lambda = 0$, $H = \text{常数}$;

$$\lambda > 0, H(\theta) = C_1 \cos \sqrt{\lambda} \theta + C_2 \sin \sqrt{\lambda} \theta$$

由于 H 周期为 2π , 则 $\sqrt{\lambda} \cdot 2\pi = 2n\pi$, $\lambda_n = n^2$

$$\therefore \lambda_n = n^2, H_n = A_n \cos n\theta + B_n \sin n\theta, (n=0, 1, 2, \dots)$$

step 2 另一种推法

设 $\bar{u}(r, \theta) = \sum_n H_n(\theta) R_n(r)$

$$r^2 \sum_{n=0}^{\infty} R_n'' \int_0^{2\pi} H_m H_n + r \sum_{n=0}^{\infty} R_n' \int_0^{2\pi} H_m H_n + \sum_{n=0}^{\infty} R_n \int_0^{2\pi} H_m H_n = 0$$

利用 $\{H_n\}$ 的 L^2 正交性, $r^2 R_n'' \int_0^{2\pi} H_n^2 + r R_n' \int_0^{2\pi} H_n^2 + R_n \int_0^{2\pi} H_n^2 = 0$

$$r^2 R_n'' + r R_n' = \lambda_n R_n \sim \text{Euler 方程}$$

$$\hat{R}_{rt} = R(e^t) \quad (\text{即令 } r = e^t)$$

$$\hat{R}_t = e^t \cdot R' = r \cdot R'$$

$$\hat{R}_{tt} = e^t \cdot R' + e^{2t} \cdot R'' = r \cdot R' + r^2 \cdot R''$$

$$\therefore \hat{R}_n'' = \lambda_n \hat{R}_n, \lambda_n = n^2, \hat{R}_{nt} = \begin{cases} a_0 + b_0 t, & n=0 \\ a_n e^{nt} + b_n e^{-nt}, & n=1, 2, 3, \dots \end{cases}$$

$$\therefore R_n(r) = \begin{cases} a_0 + b_0 \ln r, & n=0 \\ a_n r^n + b_n r^{-n}, & n=1, 2, 3, \dots \end{cases} = a_n r^n, (n=0, 1, 2, \dots)$$

由于原函数需要在 $r=0$ 时有定义, 所以 $b_n=0$ ($n=0, 1, 2, \dots$)
(or 要使 $|u|$ 有界).

$$\therefore \bar{u}(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos n\theta + B_n \sin n\theta)$$

step 3 边值条件 $\bar{u}(a, \theta) = \bar{\phi}(\theta)$ 定 A_n, B_n .

$$\bar{\phi}(\theta) = A_0 + \sum_{n=1}^{\infty} a^n (A_n \cos n\theta + B_n \sin n\theta)$$

利用 $\{\cos n\theta\}$ 和 $\{\sin n\theta\}$ 在 $L^2([0, 2\pi])$ 中的正交性.

两边积分, $A_0 = \frac{1}{2\pi} \int_0^{2\pi} \bar{\phi}(\theta) d\theta$

两边乘 $\cos n\theta$ 再积分,

$$\begin{aligned} \int_0^{2\pi} \bar{\phi}(\theta) \cos n\theta d\theta &= \int_0^{2\pi} A_0 \cos n\theta d\theta + \sum_{m=1}^{\infty} a^m \int_0^{2\pi} (A_m \cos m\theta + B_m \sin m\theta) \cdot \cos n\theta d\theta \\ &= A_n a^n \int_0^{2\pi} \cos^2 n\theta d\theta \end{aligned}$$

$$\therefore A_n = \frac{1}{\pi a^n} \int_0^{2\pi} \bar{\phi}(\theta) \cos n\theta d\theta, (n=1, 2, 3, \dots)$$

$$\text{同理得, } B_n = \frac{1}{\pi a^n} \int_0^{2\pi} \bar{\phi}(\theta) \sin n\theta d\theta, (n=1, 2, 3, \dots)$$

$$\text{代入得 } \bar{u}(r, \theta) = \frac{1}{2\pi} \int_0^{2\pi} \left[1 + 2 \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n \cos n(\tau - \theta) \right] \bar{\phi}(\tau) d\tau$$

$$u(x, y) \xrightarrow{\text{由欧拉公式化简}} \frac{1}{2\pi} \int_0^{2\pi} \left[\frac{1 - \left(\frac{r}{a}\right)^2}{1 - 2\left(\frac{r}{a}\right) \cos(\tau - \theta) + \left(\frac{r}{a}\right)^2} \right] \bar{\phi}(\tau) d\tau$$

其中 $x = r \cos \theta, y = r \sin \theta$.

: 圆盘上调和函数的泊松公式.

ex. $D = \{(x, y) : x^2 + y^2 > 1\}$, 定义在 D 上的 $u(x, y)$ 满足

$$\begin{cases} u_{xx} + u_{yy} = 0, & (x, y) \in D. \\ u(x, y) = x^2 + y & (x, y) \in \partial D \end{cases}$$

$|u|$ 是有界函数.

(1) 考虑 $v(r, \theta) = u(r \cos \theta, r \sin \theta)$. 求 $v(r, \theta)$ 所满足的方程及条件.

(2) 用分离变量法求出 $u(x, y)$.

解: (1) $v_r = u_x \cdot \cos \theta + u_y \cdot \sin \theta$, $v_\theta = u_x \cdot r(-\sin \theta) + u_y \cdot r \cdot \cos \theta$

$$\begin{aligned} v_{rr} &= \cos \theta \cdot (u_{xx} \cdot \cos \theta + u_{xy} \cdot \sin \theta) + \sin \theta \cdot (u_{xy} \cdot \cos \theta + u_{yy} \cdot \sin \theta) \\ &= u_{xx} \cdot \cos^2 \theta + u_{xy} \cdot 2 \sin \theta \cos \theta + u_{yy} \cdot \sin^2 \theta. \end{aligned}$$

$$\begin{aligned} v_{\theta\theta} &= -r \sin \theta \cdot [u_{xx} \cdot (-r \sin \theta) + u_{xy} \cdot (r \cos \theta)] + r \cos \theta [u_{xy} \cdot (-r \sin \theta) + u_{yy} \cdot r \cos \theta] \\ &\quad + u_x \cdot r \cdot (-\cos \theta) + u_y \cdot r \cdot (-\sin \theta) \end{aligned}$$

$$= u_{xx} \cdot r^2 \sin^2 \theta - u_{xy} \cdot r^2 \cdot 2 \sin \theta \cos \theta + u_{yy} \cdot r^2 \cos^2 \theta - r \cdot v_r$$

$$\therefore r^2 \cdot v_{rr} + v_{\theta\theta} + r \cdot v_r = 0$$

$$\therefore \begin{cases} v_{rr} + \frac{1}{r} \cdot v_r + \frac{1}{r^2} \cdot v_{\theta\theta} = 0, & (r, \theta) \in (1, \infty) \times [0, 2\pi) \\ |v(r, \theta)| < \infty, & (r, \theta) \in (1, \infty) \times [0, 2\pi) \\ v(1, \theta) = 2 \cos^2 \theta + \sin \theta, & \theta \in [0, 2\pi) \\ v(r, \theta) = v(r, \theta + 2\pi) \end{cases}$$

(2) 设 $v(r, \theta) = \sum_n R_n(r) H_n(\theta)$. 则有 $\frac{H''}{H} = -r^2 \cdot \frac{R''}{R} - r \cdot \frac{R'}{R} = -\lambda$

$$\begin{cases} H_n''(\theta) + \lambda_n H_n(\theta) = 0 \\ H_n(\theta) = H_n(\theta + 2\pi) \end{cases}$$

$$\therefore \lambda_n = n^2, H_n = A_n \cos n\theta + B_n \sin n\theta, \quad (n=0, 1, 2, \dots)$$

$$r^2 \cdot R_n''(r) + r \cdot R_n'(r) - \lambda_n \cdot R_n(r) = 0$$

$$\text{令 } r = e^t, \hat{R}(t) = R(e^t), \text{则 } \hat{R}''(t) = \lambda_n \cdot \hat{R}(t), \hat{R}(t) = \begin{cases} a_0 + b_0 t, & n=0 \\ a_n e^{nt} + b_n e^{-nt}, & n \geq 1 \end{cases}$$

$$\therefore R(r) = \begin{cases} a_0 + b_0 \cdot \ln r, & n=0 \\ a_n \cdot r^n + b_n \cdot r^{-n}, & n=1, 2, 3, \dots \end{cases}$$

要使 $|v(r, \theta)| < \infty$, 则 $a_n = 0 (n=1, 2, 3, \dots), b_0 = 0$.

$$r \in (1, +\infty), R(r) \text{ 有界.}$$

$$\therefore v(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^{-n} (A_n \cos n\theta + B_n \sin n\theta)$$

$$v(1, \theta) = A_0 + \sum_{n=1}^{\infty} (A_n \cos n\theta + B_n \sin n\theta) = 2 \cos^2 \theta + \sin \theta$$

$$= 1 + \sin \theta + \cos 2\theta$$

$$\therefore A_0 = 1, B_1 = 1, A_2 = 1, A_n = 0 (n=1, 3, 4, \dots), B_n = 0 (n=2, 3, 4, \dots)$$

$$\therefore v(r, \theta) = 1 + \frac{\sin \theta}{r} + \frac{\cos 2\theta}{r^2}$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \Rightarrow \begin{cases} r = \sqrt{x^2 + y^2} \\ \sin \theta = \frac{y}{\sqrt{x^2 + y^2}}, \cos \theta = \frac{x}{\sqrt{x^2 + y^2}} \end{cases}$$

$$\therefore u(x, y) = 1 + \frac{y}{x^2 + y^2} + \frac{2 \cdot \frac{x^2}{x^2 + y^2} - 1}{x^2 + y^2}$$

$$= 1 + \frac{y}{x^2 + y^2} + \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

ex. $D = \{(x, y) : x^2 + y^2 > 1, y \geq 0\}$.

区域 $\rightarrow$ 不一样的地方
边值条件 $\rightarrow$

$$\begin{cases} u_{xx} + u_{yy} = 0, (x, y) \in D \\ u|_{x^2+y^2=1, y \geq 0} = 4xy, u|_{y=0, |x| \geq 1} = 0, \\ |u| \text{ 是有界函数.} \end{cases}$$

(1) $v(r, \theta) = u(r \cos \theta, r \sin \theta)$ 满足 $\begin{cases} v_{rr} + \frac{1}{r} v_r + \frac{1}{r^2} v_{\theta\theta} = 0, \\ |v(r, \theta)| < \infty, \\ v(1, \theta) = 4 \cos \theta \sin \theta, \theta \in [0, \pi) \\ v(r, 0) = v(r, \pi) = 0. \end{cases} (r, \theta) \in [1, \infty) \times [0, \pi)$

(2) $H'' + \lambda H = 0$ $\lambda_n = n^2, H_n(\theta) = A_n \cdot \sin n\theta, (n=1, 2, 3, \dots)$
 $\begin{cases} H(0) = H(\pi) = 0 \end{cases}$

$$r^2 \cdot R'' + r \cdot R' = \lambda_n \cdot R, R_n(r) = \begin{cases} a_0 + b_0 \cdot \ln r, & n=0 \\ a_n \cdot r^n + b_n \cdot r^{-n}, & n=1, 2, 3, \dots \end{cases}, |v(r, \theta)| < \infty, b_0 = 0, a_n = 0.$$

$$\therefore v(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^{-n} \cdot A_n \cdot \sin n\theta, v(1, \theta) = 2 \sin 2\theta, A_0 = 0, A_2 = 2, A_n = 0 (n=1, 3, 4, \dots)$$

$$\therefore v(r, \theta) = r^{-2} \cdot 2 \sin 2\theta, u(x, y) = \frac{4xy}{(x^2 + y^2)^2}$$

§3.2 非齐次方程、齐次边值条件的定解问题

齐次边值的热传导方程: Duhamel 方法 → 可用于非齐次方程, 非齐热方程.

$$\begin{cases} \partial_t u = a^2 \partial_x^2 u + f(x, t) & , (x, t) \in (0, l) \times (0, \infty) \\ u(x, 0) = \phi(x) & , x \in (0, l) \\ u(0, t) = 0, u(l, t) = 0 & , t \in (0, \infty) \end{cases}$$

线性叠加: $u(x, t) = u_0(x, t) + u_1(x, t)$

其中 $u_0(x, t)$ 是齐次方程、非齐初值问题的解:

$$\begin{cases} \partial_t u_0 = a^2 \partial_x^2 u_0 & , (x, t) \in (0, l) \times (0, \infty) \\ u_0(x, 0) = \phi(x) & , x \in (0, l) \\ u_0(0, t) = 0, u_0(l, t) = 0 & , t \in (0, \infty) \end{cases}$$

而 $u_1(x, t)$ 是非齐次方程、齐次初值问题的解:

$$(\star) \begin{cases} \partial_t u_1 = a^2 \partial_x^2 u_1 + f(x, t) & , (x, t) \in (0, l) \times (0, \infty) \\ u_1(x, 0) = 0 & , x \in (0, l) \\ u_1(0, t) = 0, u_1(l, t) = 0 & , t \in (0, \infty) \end{cases}$$

由分离变量法, $u_0(x, t) = \sum_{n=1}^{\infty} A_n e^{-(\frac{n\pi a}{l})^2 t} \sin \frac{n\pi}{l} x$, $A_n = \frac{2}{l} \int_0^l \phi(s) \sin \frac{n\pi}{l} s ds$

对 $(\star)$ 使用 Duhamel 原理: 考虑 $w(x, t; \tau)$ 满足

$$\begin{cases} \partial_t w = a^2 \partial_x^2 w & , (x, t) \in (0, l) \times (0, \infty) \\ w(x, 0; \tau) = f(x, \tau) & , x \in (0, l) \\ w(0, t; \tau) = 0, w(l, t; \tau) = 0 & , t \in (0, \infty) \end{cases}$$

由分离变量法, $w(x, t; \tau) = \sum_{n=1}^{\infty} B_n(\tau) e^{-(\frac{n\pi a}{l})^2 t} \sin \frac{n\pi}{l} x$,

$$B_n(\tau) = \frac{2}{l} \int_0^l f(s, \tau) \sin \frac{n\pi}{l} s ds$$

可以验证 $u_1(x, t) = \int_0^t w(x, t-\tau; \tau) d\tau$ Leibniz 积分法则

$$\therefore u_1(x, t) = \frac{2}{l} \sum_{n=1}^{\infty} \left(\sin \frac{n\pi}{l} x \right) \int_0^t \left[\int_0^l f(s, \tau) \sin \frac{n\pi}{l} s ds \right] e^{-(\frac{n\pi a}{l})^2 (t-\tau)} d\tau$$

另例: 齐次边值的热方程: 特征函数展开法

$$\begin{cases} \partial_t u = a^2 \partial_x^2 u + f(x, t), & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \phi(x), & x \in (0, L) \\ u(0, t) = 0, u_x(L, t) = 0, & t \in (0, \infty) \end{cases}$$

step 1: 特征值, 特征函数

$$\begin{cases} X''(x) + \lambda X(x) = 0, & 0 < x < L \\ X(0) = X'(L) = 0 \end{cases}$$

得 $\lambda_n = \frac{(n - \frac{1}{2})^2 \pi^2}{L^2}$, $X_n = \sin \frac{(n - \frac{1}{2}) \pi x}{L}$, $n = 1, 2, 3, \dots$

step 2: 设 $u(x, t) = \sum_{m=1}^{\infty} X_m(x) T_m(t)$

$$\sum_{m=1}^{\infty} \int_0^L X_m^2 T_m' = a^2 \sum_{m=1}^{\infty} \int_0^L \underbrace{X_m X_m''}_{-\lambda X_m} T_m + \int_0^L f \cdot X_n$$

$$\int_0^L X_n^2 \cdot T_n' = -\lambda_n a^2 \cdot T_n \cdot \int_0^L X_n^2 + \int_0^L f \cdot X_n dx$$

$$\Rightarrow T_n' = -\lambda_n a^2 T_n + \frac{\int_0^L f \cdot X_n dx}{\int_0^L X_n^2} \rightarrow f_n(t)$$

$$\int_{X_n} \phi(x) = u(x, 0) = \sum_{m=1}^{\infty} \int_{X_n(x)} X_m(x) T_m(0) = T_n(0) \int_0^L X_n^2$$

齐次边值的弦振动方程: 特征函数展开法

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f(x, t), & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), & x \in (0, L) \\ u_x(0, t) = 0, u_x(L, t) = 0, & t \in (0, \infty). \end{cases}$$

step 1: $\begin{cases} X'' + \lambda X = 0, & 0 < x < L \\ X'(0) = X'(L) = 0 \end{cases} \Rightarrow \lambda_n = \frac{n^2 \pi^2}{L^2}, X_n = \cos \frac{n \pi x}{L}, n = 0, 1, 2, \dots$

step 2: $u(x, t) = \sum_n X_n(x) T_n(t)$

$$\begin{cases} T_n''(t) + a^2 \lambda_n T_n(t) = f_n(t) \\ T_n(0) = \phi_n, T_n'(0) = \psi_n \end{cases}$$

$$f_0(t) = \frac{1}{l} \int_0^l f(s, t) ds, \quad f_n(t) = \frac{2}{l} \int_0^l f(x, t) \cos \frac{n\pi x}{l} dx$$

$$\phi_0 = \frac{1}{l} \int_0^l \phi(s) ds, \quad \phi_n = \frac{2}{l} \int_0^l \phi(s) \cos \frac{n\pi s}{l} ds \quad (n=1, 2, 3, \dots)$$

$$\psi_0 = \frac{1}{l} \int_0^l \psi(s) ds, \quad \psi_n = \frac{2}{l} \int_0^l \psi(s) \cos \frac{n\pi s}{l} ds$$

$$T_n(t) = \begin{cases} \phi_0 + \psi_0 t + \int_0^t f_0(\tau) (t-\tau) d\tau, & n=0 \\ \phi_n \cos \frac{n\pi a}{l} t + \psi_n \sin \frac{n\pi a}{l} t + \frac{l}{n\pi a} \int_0^t f_n(\tau) \sin \frac{n\pi a}{l} (t-\tau) d\tau, & n \geq 1 \end{cases}$$

Duhamel 原理的非齐次常微分方程:

$$\begin{cases} T''(t) + bT'(t) + cT(t) = f(t), & 0 < x < l \\ T(0) = \phi_0, T'(0) = \psi_0 \end{cases}$$

线性性: $T(t) = y(t) + z(t)$. 其中 $\begin{cases} y''(t) + by'(t) + cy(t) = 0 \\ y(0) = \phi_0, y'(0) = \psi_0 \end{cases} \quad \begin{cases} z''(t) + bz'(t) + cz(t) = f(t) \\ z(0) = 0, z'(0) = 0 \end{cases}$

为求 z , 考虑如下 $w(t; \tau)$: $\begin{cases} w''(t; \tau) + bw'(t; \tau) + cw(t; \tau) = 0 \\ w(0; \tau) = 0, w'(0; \tau) = f(\tau) \end{cases}$

$$z(t) = \int_0^t w(t-\tau; \tau) d\tau$$

[3.4] 常数变易法时 = 阶常系数线性 ODE

$$y''(x) + by'(x) + cy(x) = f(x)$$

$$y = y_h + y_p$$

homogeneous particular

$$y_h = C_1 y_1 + C_2 y_2$$

$$\Delta = b^2 - 4c$$

$$\Delta > 0$$

$$\Delta = 0$$

$$\Delta < 0$$

$$\text{特征方程 } \mu^2 + b\mu + c = 0$$

$$\mu_1, \mu_2$$

$$\mu (= \text{重根})$$

$$\mu_1, \mu_2 = \alpha \pm i\beta$$

齐次通解

$$y_1 = e^{\mu_1 x}, y_2 = e^{\mu_2 x}$$

$$y_1 = e^{\mu x}, y_2 = x e^{\mu x}$$

$$y_1 = e^{\alpha x} \cos(\beta x), y_2 = e^{\alpha x} \sin(\beta x)$$

常数变易法: $y_p = u_1(x) y_1 + u_2(x) y_2$

$$y_p' = u_1 y_1' + u_2 y_2' \quad (\text{令 } \underline{u_1' y_1 + u_2' y_2 = 0})$$

$$y_p'' = u_1 y_1'' + u_2 y_2'' + u_1' y_1' + u_2' y_2'$$

$$\therefore \underline{u_1' y_1' + u_2' y_2' = f},$$

$$\text{得 } u_1 = - \int \frac{y_2 f}{W}, u_2 = \int \frac{y_1 f}{W}$$

$$\text{其中 } W = y_1 y_2' - y_1' y_2 \quad (\text{Wronskian})$$

矩形上泊松方程: 特征函数展开法

$$\begin{cases} u_{xx} + u_{yy} = 1, (x, y) \in (0, 2)^2 \\ u(0, y) = 0, u(2, y) = \cos^2 y \\ u_y(x, 0) = u_y(x, 2) = 0. \end{cases}$$

$$\text{step 1: } \begin{cases} Y''(y) + \lambda Y(y) = 0 \quad (0 < y < 2) \\ Y'(0) = Y'(2) = 0. \end{cases} \Rightarrow \lambda_n = n^2, Y_n = \cos ny, n = 0, 1, 2, 3, \dots$$

$$\text{step 2: } u(x, y) = \sum_n X_n(x) Y_n(y), \quad \begin{cases} X_n''(x) - \lambda_n X_n(x) = f_n \quad (0 < x < 2) \\ X_n(0) = \phi_n, X_n(2) = \psi_n. \end{cases} \quad (\star)$$

其中 f_n, ϕ_n, ψ_n 分别为: $f_0 = 1, f_n = 0 \ (n \geq 1)$

$$\phi_n = 0 \ (n \geq 0)$$

$$\psi_0 = \psi_2 = \frac{1}{2}, \psi_n = 0 \ (n \neq 0, 2)$$

ex.
$$\begin{cases} (t+1)u_t = u_{xx} + x, & (x, t) \in (0, 2) \times (0, \infty) \\ u(0, t) = u(2, t) = 0, & t \in [0, \infty) \\ u(x, 0) = \sin x, & x \in [0, 2] \end{cases}$$

step 1.
$$\begin{cases} X'' + \lambda X = 0, & 0 < x < 2 \\ X(0) = X(2) = 0 \end{cases} \Rightarrow \lambda_n = n^2, X_n(x) = \sin nx, n = 1, 2, 3, \dots$$

step 2. 设 $u(x, t) = \sum_n X_n(x) T_n(t)$, 则有
$$\begin{cases} (t+1)T_n' + n^2 T_n = f_n & (t > 0) \\ T_n(0) = \phi_n \end{cases}$$

其中 $f_n = \frac{2}{\pi} \int_0^2 x \sin nx \, dx = (-1)^{n-1} \cdot \frac{2}{n}$

$$\phi_1 = 1, \phi_n = 0 \ (n > 1)$$

§ 3.3 非齐次方程、非齐次边值条件的定解问题是

第 I 类边值问题: 边值齐次化

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f(x, t) & (x, t) \in (0, l) \times (0, \infty) \\ u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), & x \in (0, l) \\ u(0, t) = \gamma(t), u(l, t) = \mu(t) & t \in (0, \infty) \end{cases}$$

w 的取法很多, 满足需要即可.

取 $w(x, t) = \frac{l-x}{l} \gamma(t) + \frac{x}{l} \mu(t)$. 满足 $w(0, t) = \gamma(t), w(l, t) = \mu(t)$

考虑 $v(x, t) = u(x, t) - w(x, t)$. 满足:

$$\begin{cases} \partial_t^2 v = a^2 \partial_x^2 v + f(x, t) - \partial_t^2 w \\ v(x, 0) = \phi(x) - w(x, 0), \partial_t v(x, 0) = \psi(x) - \partial_t w(x, 0) \\ v(0, t) = 0, v(l, t) = 0 \end{cases}$$

可以用特征函数展开法得到 v , 从而得到 u .

第II类边值问题: 边值齐次化

$$\begin{cases} \partial_t u = a^2 \partial_x^2 u + f(x, t), & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \phi(x), & x \in (0, L) \\ u_x(0, t) = \gamma(t), u_x(L, t) = \mu(t), & t \in (0, \infty) \end{cases}$$

取 $w(x, t) = (x - \frac{x^2}{2L})\gamma(t) + \frac{x^2}{2L}\mu(t)$. 满足 $w_x(0, t) = \gamma(t)$, $w_x(L, t) = \mu(t)$

考虑 $v(x, t) = u(x, t) - w(x, t)$.

若边值条件为 $u(0, t) = \gamma(t)$, $u_x(L, t) = \mu(t)$. 取 $w(x, t) = \gamma(t) + x \cdot \mu(t)$

$$u_x(0, t) = \gamma(t), u(L, t) = \mu(t), \quad w(x, t) = \mu(t) - \gamma(t)(L-x)$$

特殊的第I类边值问题: 边值齐次化.

有界弦的强迫振动

$$\begin{cases} \partial_t^2 u = a^2 \partial_x^2 u + f(x), & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \phi(x), \partial_t u(x, 0) = \psi(x), & x \in (0, L) \\ u(0, t) = A, u(L, t) = B, & t \in (0, \infty) \end{cases}$$

$$v(x, t) = u(x, t) - w(x).$$

$$\begin{cases} \partial_t^2 v = a^2 \partial_x^2 v + f(x) + a^2 w''(x), & (x, t) \in (0, L) \times (0, \infty) \\ v(x, 0) = \phi(x) - w(x), \partial_t v(x, 0) = \psi(x), & x \in (0, L) \\ v(0, t) = A - w(0), v(L, t) = B - w(L), & t \in (0, \infty) \end{cases}$$

$$\text{取 } w \text{ 是 } \begin{cases} w''(x) = -\frac{f(x)}{a^2}, & x \in (0, L) \\ w(0) = A, w(L) = B \end{cases} \text{ 的解}$$

则 v 满足齐次方程, 齐次边值问题